

LAB-4399: From VMware to OpenShift Virtualization

A Hands-On Migration Lab

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STSM, IBM Cloud Solution Architecture

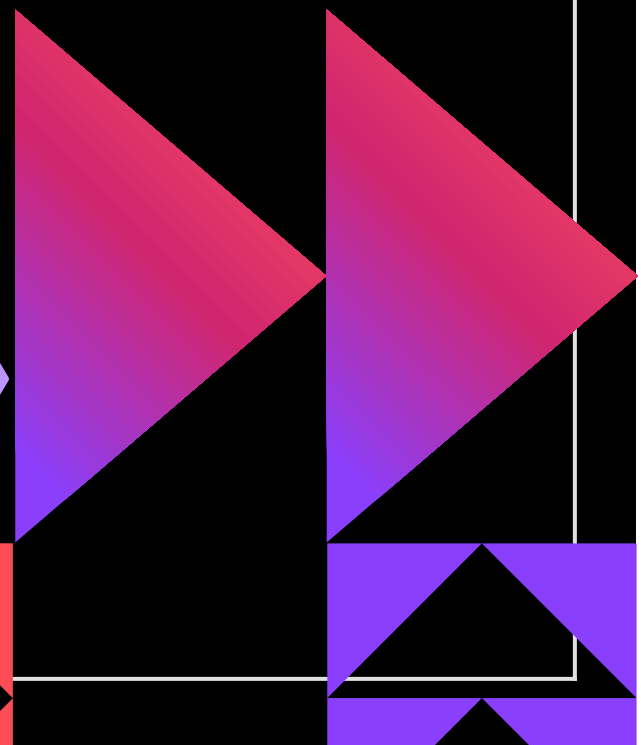
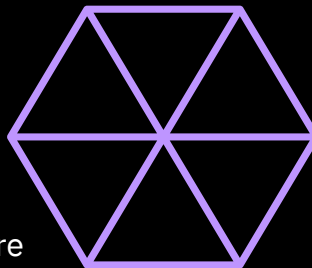


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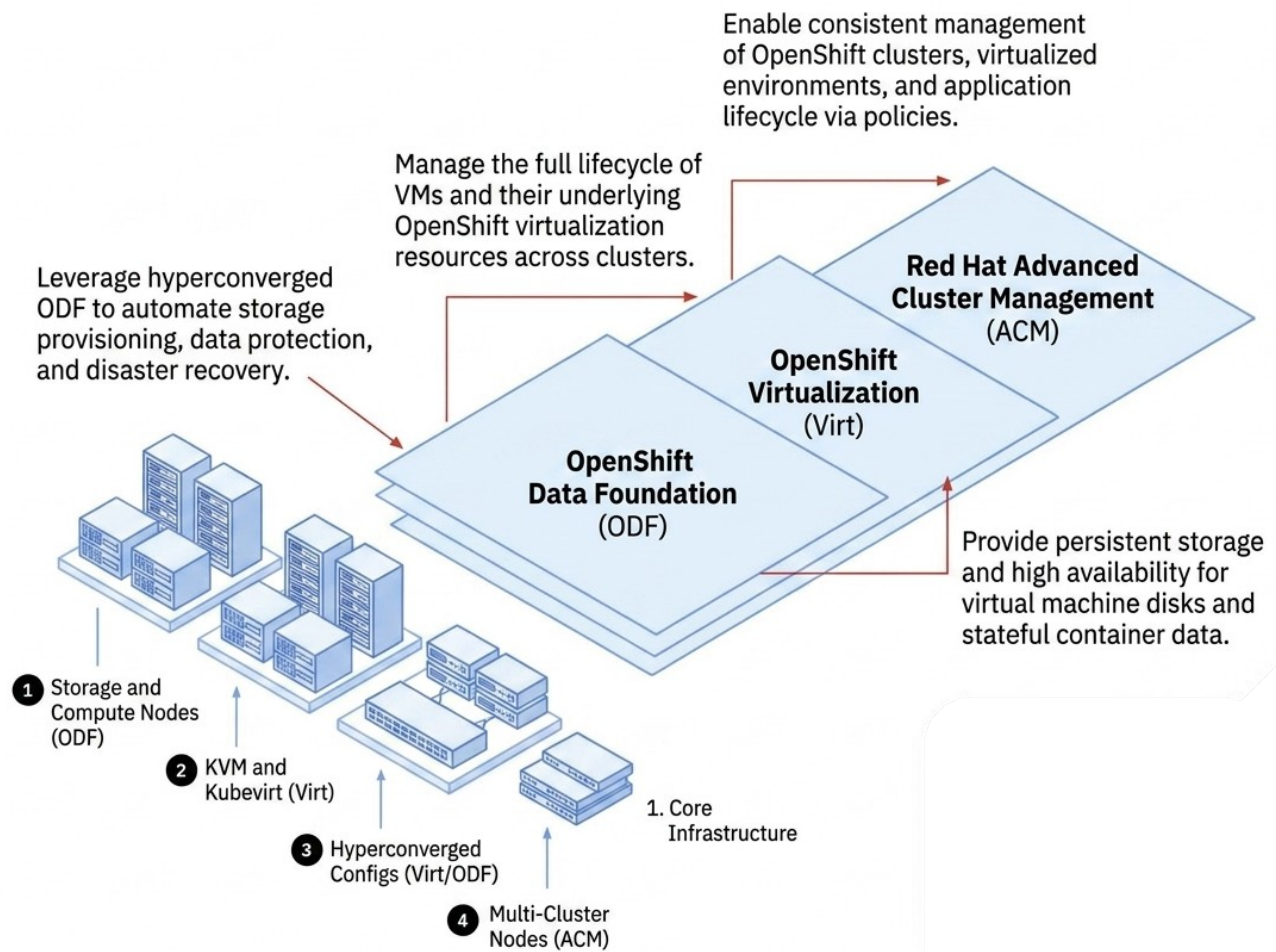
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Introduction



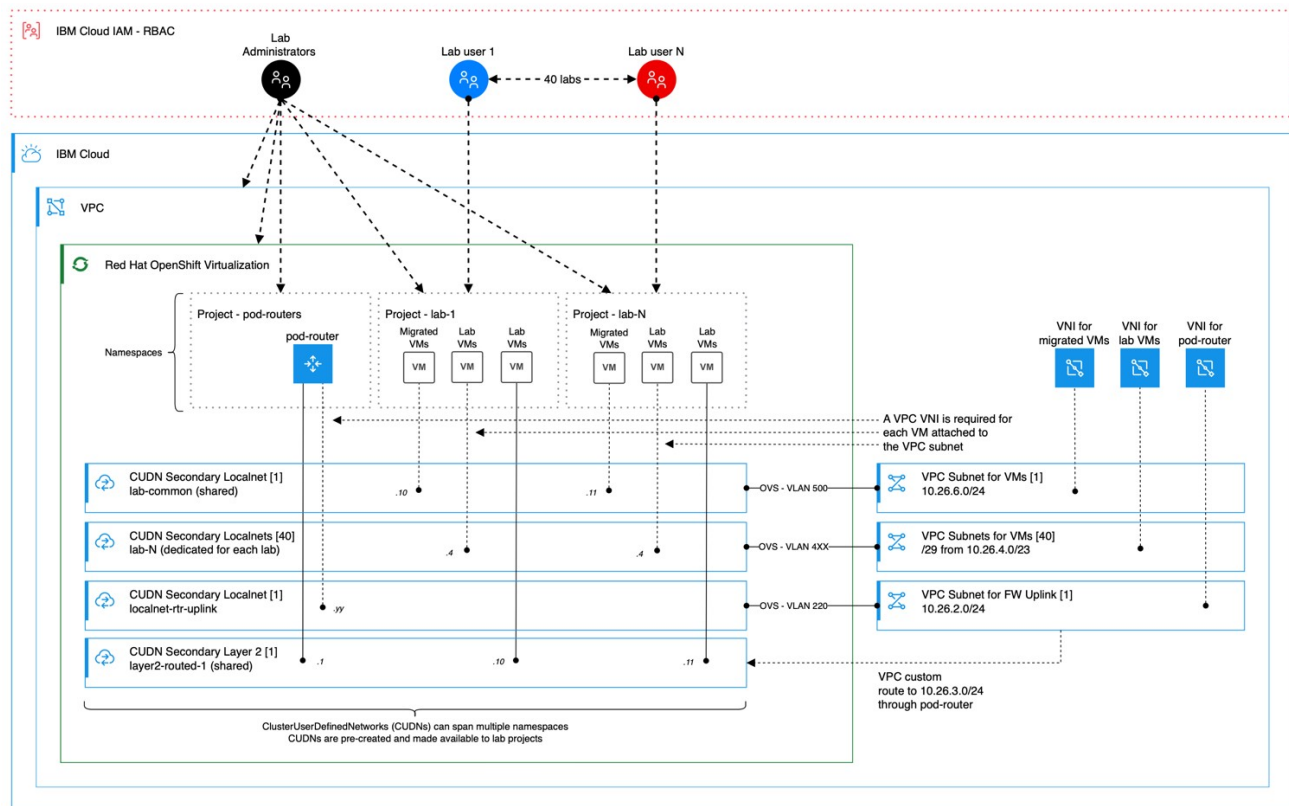
Virtualization modernization is a key step in helping organization increase flexibility, reduce complexity, and prepare for future application modernization initiatives. OpenShift Virtualization enable enterprises to run existing virtual machines alongside containers on a single platform, simplifying operations and accelerating innovation. In this hands-on lab, you will learn how to migrate VMware workloads to OpenShift Virtualization and experience the technologies that help organizations modernize their virtualization environments with confidence.

Demo

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Prerequisites:

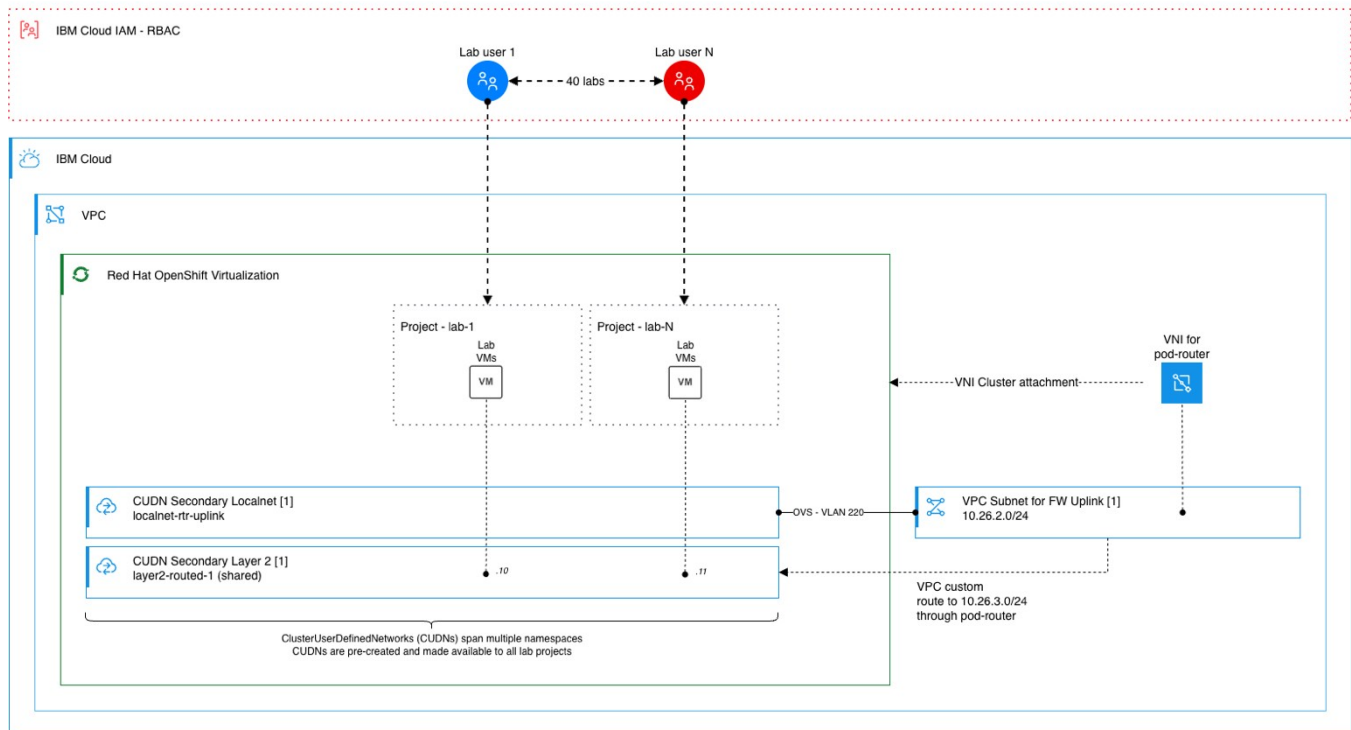
- A working **ROVS cluster** with all operators healthy.
- The **oc**, **virtctl**, and **ibmcloud** CLIs installed.
- A **VPC subnet** with a Public Gateway attached (for outbound internet access).
- At least one allowed **VLAN** on your bare metal server's PCI network attachment.
- **OVS bridge** and bridge mapping NNCPs applied and available.



Provision a VM using Layer 2 Primary UDN

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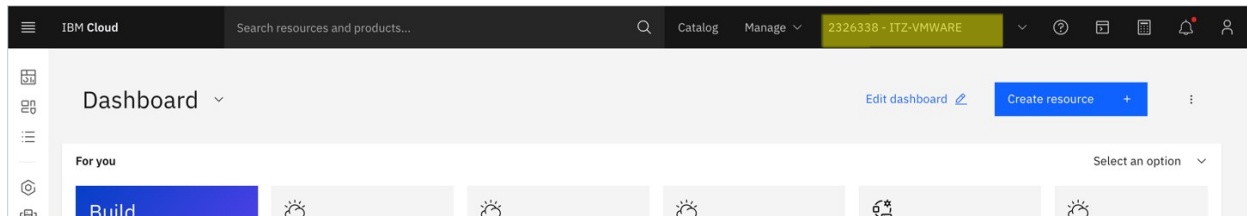
A VM whose entire pod network is a Layer 2 primary CUDN. No VNI, no VLAN, no MAC pinning. IP is DHCP-assigned from the CUDN subnet and persistent for the life of the VM.



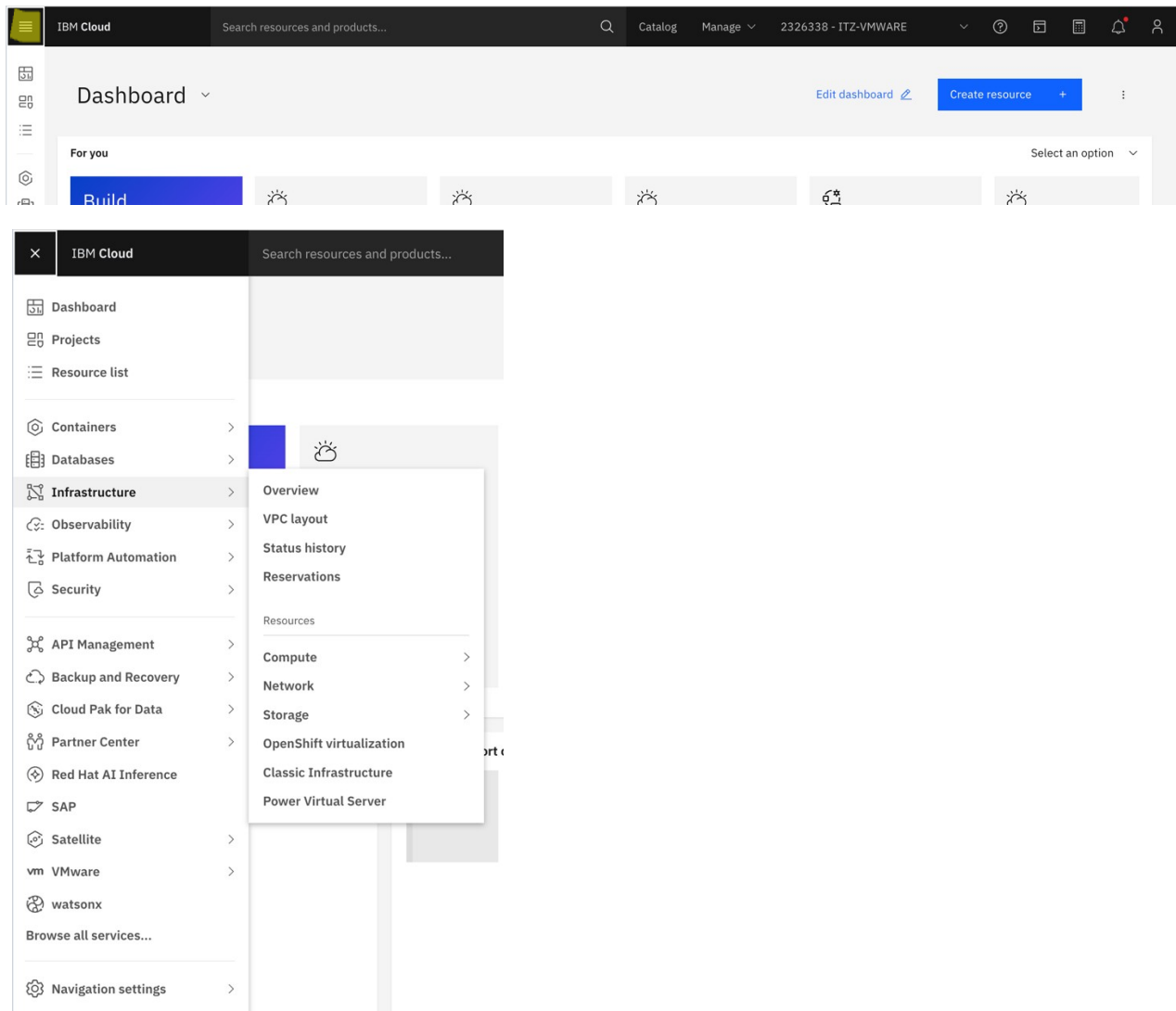
1. Access the OpenShift Web Console

- Open cloud.ibm.com and confirm the account selector (upper right) is set to **2326338 – ITZ-VMWARE**.

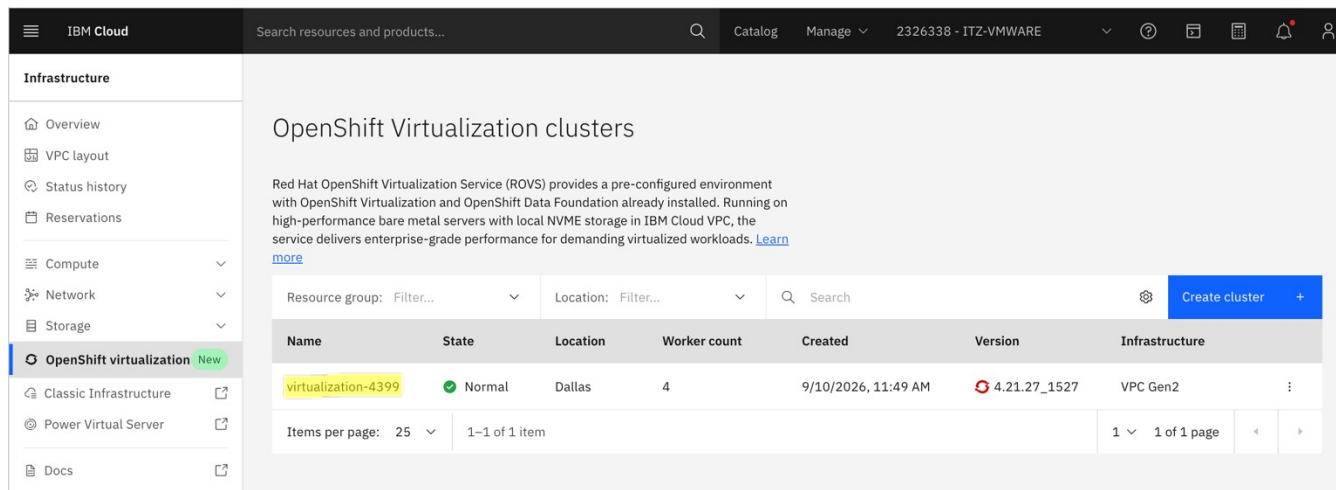
IBM TechXchange



b. Click the hamburger menu (upper left) → **Infrastructure** → **OpenShift virtualization**.



c. Click the cluster name **virtualization-4399**.



OpenShift Virtualization clusters

Red Hat OpenShift Virtualization Service (ROVS) provides a pre-configured environment with OpenShift Virtualization and OpenShift Data Foundation already installed. Running on high-performance bare metal servers with local NVME storage in IBM Cloud VPC, the service delivers enterprise-grade performance for demanding virtualized workloads. [Learn more](#)

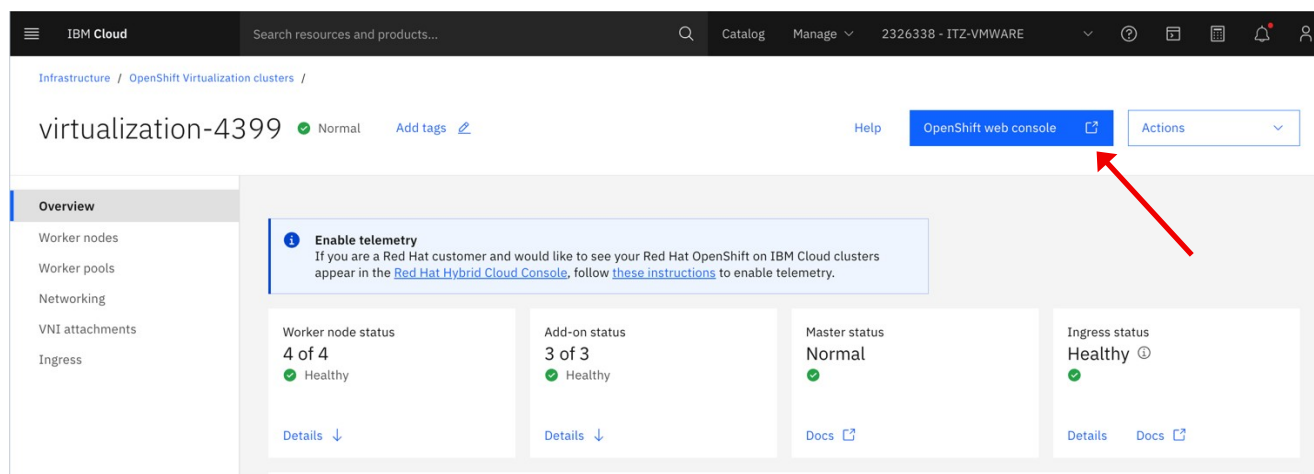
Resource group: Filter... Location: Filter... Search

Create cluster +

Name	State	Location	Worker count	Created	Version	Infrastructure
virtualization-4399	Normal	Dallas	4	9/10/2026, 11:49 AM	4.21.27_1527	VPC Gen2

Items per page: 25 1–1 of 1 item 1 1 of 1 page

d. Click the blue **OpenShift web console** button (upper right of the cluster detail page).



virtualization-4399 Normal Add tags

Help OpenShift web console Actions

Enable telemetry
If you are a Red Hat customer and would like to see your Red Hat OpenShift on IBM Cloud clusters appear in the [Red Hat Hybrid Cloud Console](#), follow [these instructions](#) to enable telemetry.

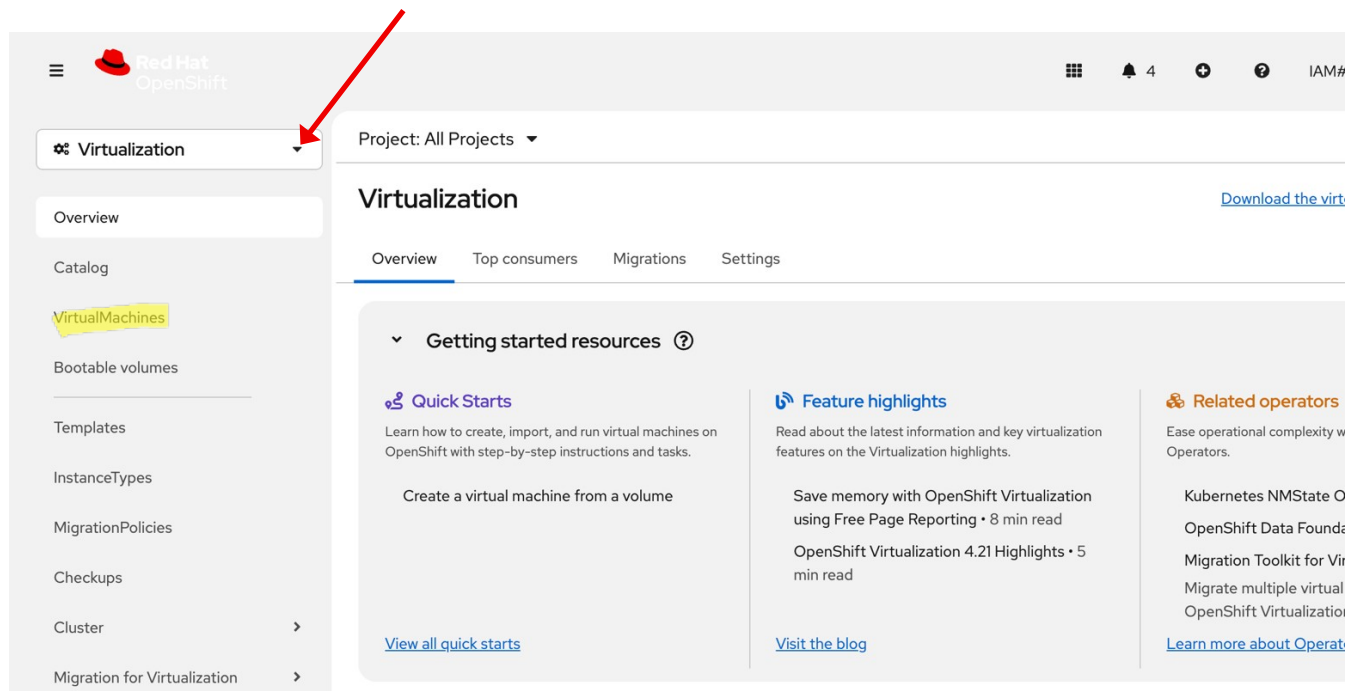
Worker node status	Add-on status	Master status	Ingress status
4 of 4 Healthy	3 of 3 Healthy	Normal	Healthy

Details Docs

The OpenShift console opens in a new tab, already authenticated.

2. Navigate to Virtualization

- In the drop down, click **Virtualization** then **VirtualMachines**. Leave the namespace set to **All Namespaces** — no need to find your namespace in the dropdown.



3. Create a Virtual Machine from Instance Type

- a. Click **Create** → **From Instance Type**

Summary

All projects

0 Projects

Virtual Machines (0)

0 Error

0 Running

0 Stopped

0 Other

Usage

CPU	Memory	Storage
0 m	0 B	0 B
Requested of 0 m	Used of 0 B	Used of 0 B



No VirtualMachines found

Click **Create VirtualMachine** to create your first VirtualMachine or view the [catalog](#) tab to create a VirtualMachine from the available options

Create VirtualMachine

From InstanceType

From template

With YAML

4. Create the VM

- Select **centos-stream9**.

1 Select volume to boot from

Add volume

Volumes project

PR All projects

Filter

Search by name...

1 - 8 of 9

Show all

★ ↑	Volume name ↓	Architecture	Namespace ↓	Operating system ↓	Storage class ↓	Size ↑...	Description ↓...
★	centos-stream9	-	openshift-virtualization-os-images	CentOS Stream 9	ocs-storagecluster-rbdplugin-snapclass	30 GiB	-

- Scroll down and select **General Purpose**. Click the **U series** drop down and select **small: 1CPUs, 2 GiB Memory**.

2 Select InstanceType

Red Hat provided User provided

Network N series

Overcommitted O series

Memory Intensive M series

Realtime RT series

Dedicated vCPU D series

Compute Exclusive CX series

U series

small: 1 CPU, 2 GiB Memory

c. Scroll down and change the vm name to **<initials>-layer2-vm** (ex: js-layer2-vm).

3 VirtualMachine details

Name * js-layer2-vm

Operating system CentOS Stream 9

InstanceType u1.small

CPU | Memory 1 CPU | 2 GiB Memory

Project lab-4399-ph-lab-1

Disk size 30 GiB

Storage class ocs-storagecluster-ceph-rbd-virtualization

Public SSH key Not configured

Dynamic SSH key injection

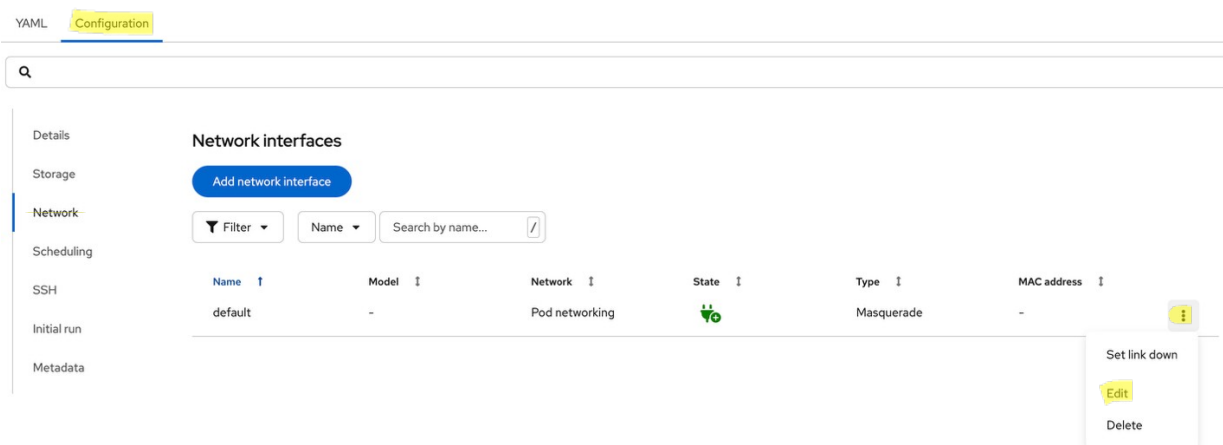
d. **DO NOT CLICK CREATE.** Click **Customize VirtualMachine.**

☒ Start this VirtualMachine after creation

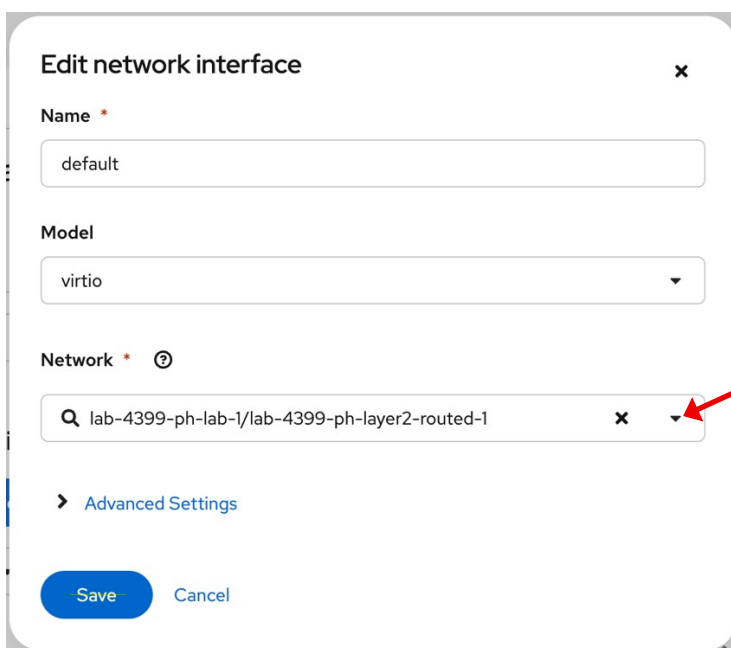
Create VirtualMachine Customize VirtualMachine Cancel

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- e. In the Customize screen, click the **Configuration** tab, then navigate to the **Network** tab on the left. Click the kebab menu (:) on the **default** interface row and choose **Edit**.



- f. In the Network dropdown, select the **Layer-2** network binding type. Click **Save**.



- g. Click **Create VirtualMachine**.

☒ Start this VirtualMachine after creation



5. Watch the VM start

- Watch the status column update from **Provisioning** → **Starting** → **Running**.

VM js-layer2-vm2 Running

Overview Metrics YAML Configuration Events Console Snapshots Diagnostics

Details

Name	js-layer2-vm2 amd64	VNC console Open web console
Status	Running	
Created	2 minutes ago (2 minutes ago)	
Operating system	CentOS Stream 9	
CPU Memory	1 CPU 2 GiB Memory	
Time zone	EDT	
InstanceType	CR u1.small	
Preference	CR centos.stream9	
Hostname	js-layer2-vm2	

- View the Network details.

Network (1)

Name	IP address
default	10.26.3.15
Internal FQDN	js-layer2-vm2.headless.lab-...

6. Open the In-Browser Console

- a. In the VM Details section, click **Open web console**.

[Details](#)

Name	js-layer2-vm2 amd64	VNC console
Status	Running	Open web console
Created	6 minutes ago (6 minutes ago)	
Operating system	CentOS Stream 9	
CPU Memory	1 CPU 2 GiB Memory	

- b. Login in with the guest login credentials. Then run: `ip addr show`

Hint: Use the copy and Paste to console buttons.

Guest login credentials ⓘ
User name
Password

Paste to console en-US

```

CentOS Stream 9
Kernel 5.14.0-749.el9.x86_64 on an x86_64

Activate the web console with: systemctl enable --now cockpit.socket

js-layer2-vm2 login: centos
Password:
[centos@js-layer2-vm2 ~]$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 8900 qdisc fq_codel state UP group default qlen 1000
    link/ether 02:a1:c3:6e:18:8d brd ff:ff:ff:ff:ff:ff
    altname enp1s0
    inet 10.26.3.15/24 brd 10.26.3.255 scope global dynamic noprefixroute eth0
        valid_lft 42564sec preferred_lft 42564sec
    inet6 fe80::a1:c3:ff:fe:18:8d/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[centos@js-layer2-vm2 ~]$ _

```

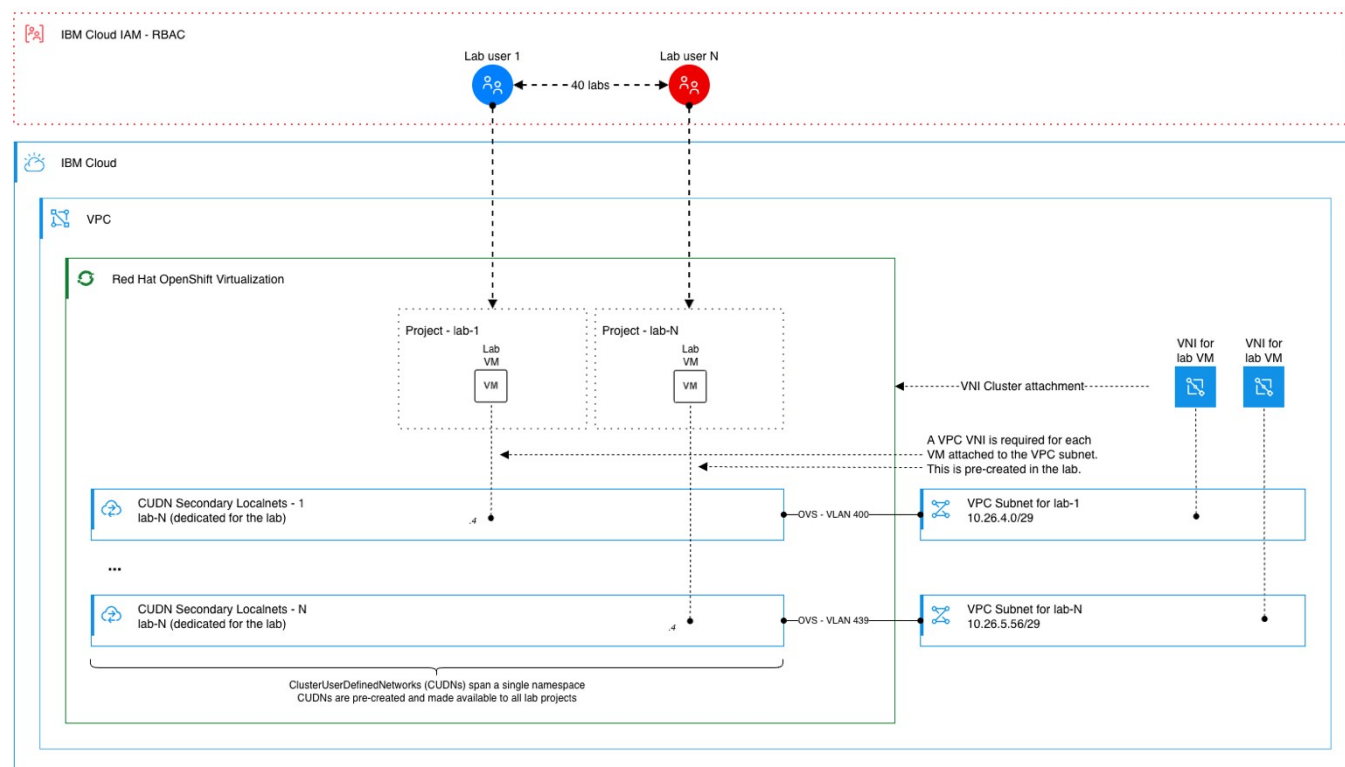
- c. Then run: `curl http://lab.techzone.ibm.local/`

Congratulations, you have completed part 1 of this lab!

Provision a New VM – Localnet UDN

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Localnet networking is the most “like for like” network setup when compared to on-premise deployments of Red Hat OpenShift Virtualization Engine. Localnet is the simplest and easiest to understand, but it may not be the most effective way to scale deployments of VMs across ROVS.

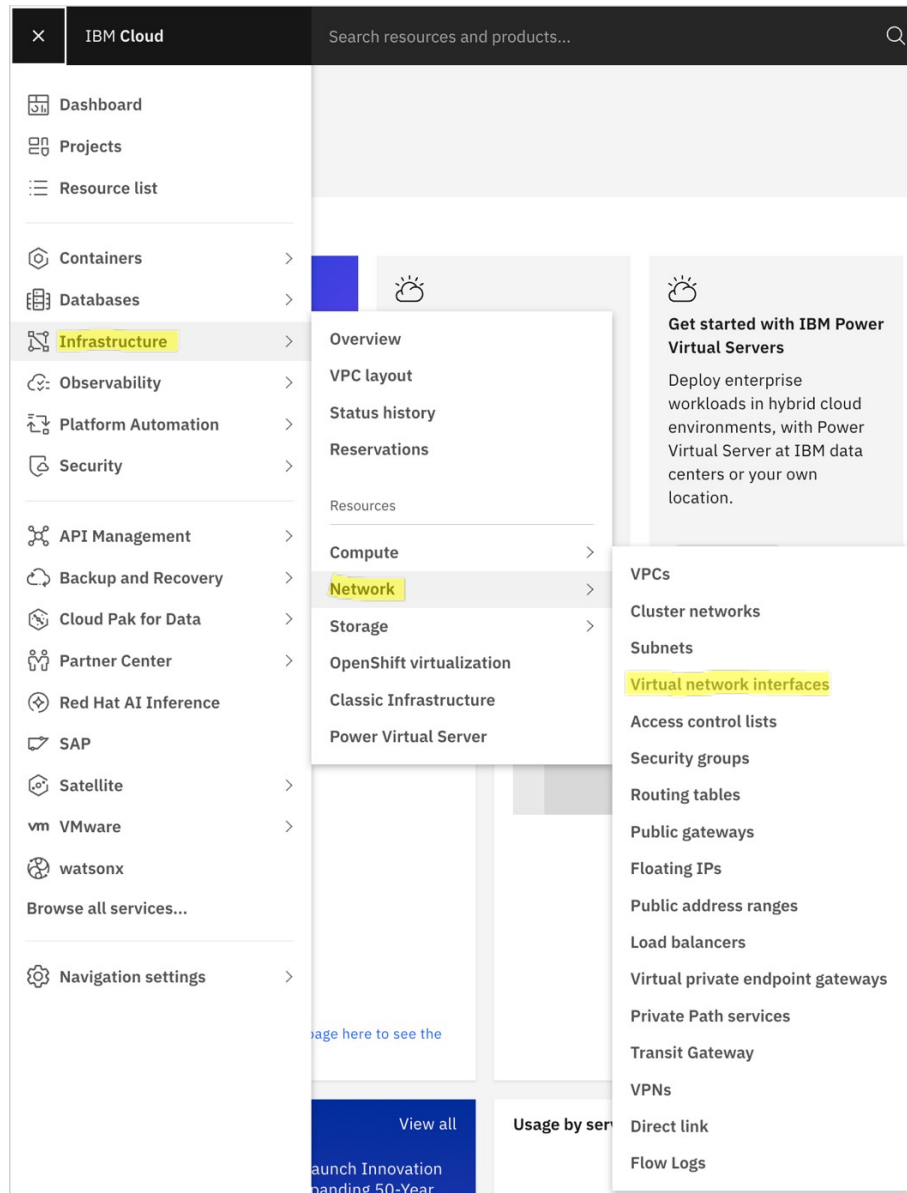


1. Create the Virtual Network Interface (VNI)

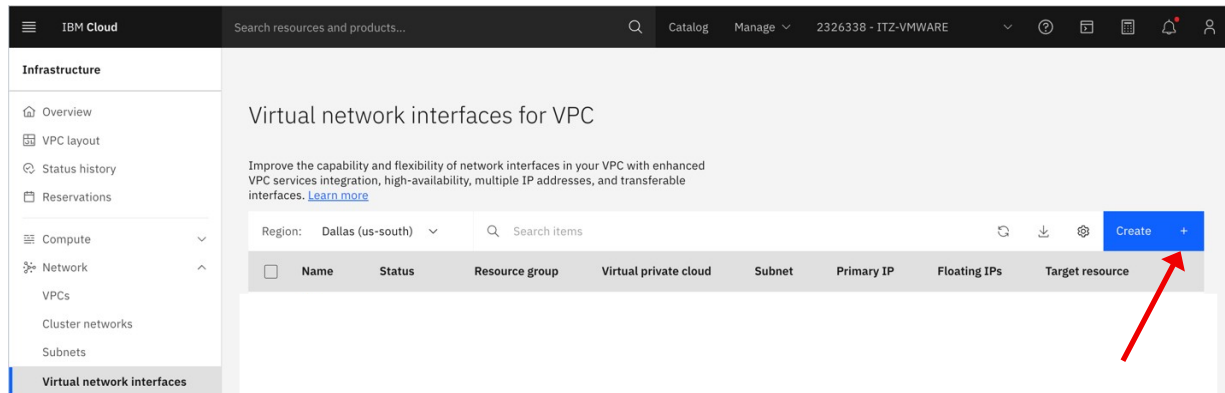
We are going to walk through the process of creating a VNI, but **DO NOT CLICK CREATE** at the end, as there is already a VNI created and attached for you.

- b. Click the hamburger menu → **Infrastructure** → **Network** → **Virtual network interfaces**.

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b. Click **Create**.



b. Fill in the form:

- **Region:** Dallas (us-south) – confirm, do not change
- **Name:** lab-4399-example
- **Resource group:** techxchange-2026-lab-4399

Infrastructure / Virtual network interfaces /

Virtual network interface for VPC

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Create About

Location ⓘ
Select the location where you want to create your virtual network interface.

Region

Details

Name

Use lowercase alphanumeric characters and hyphens only (without spaces).

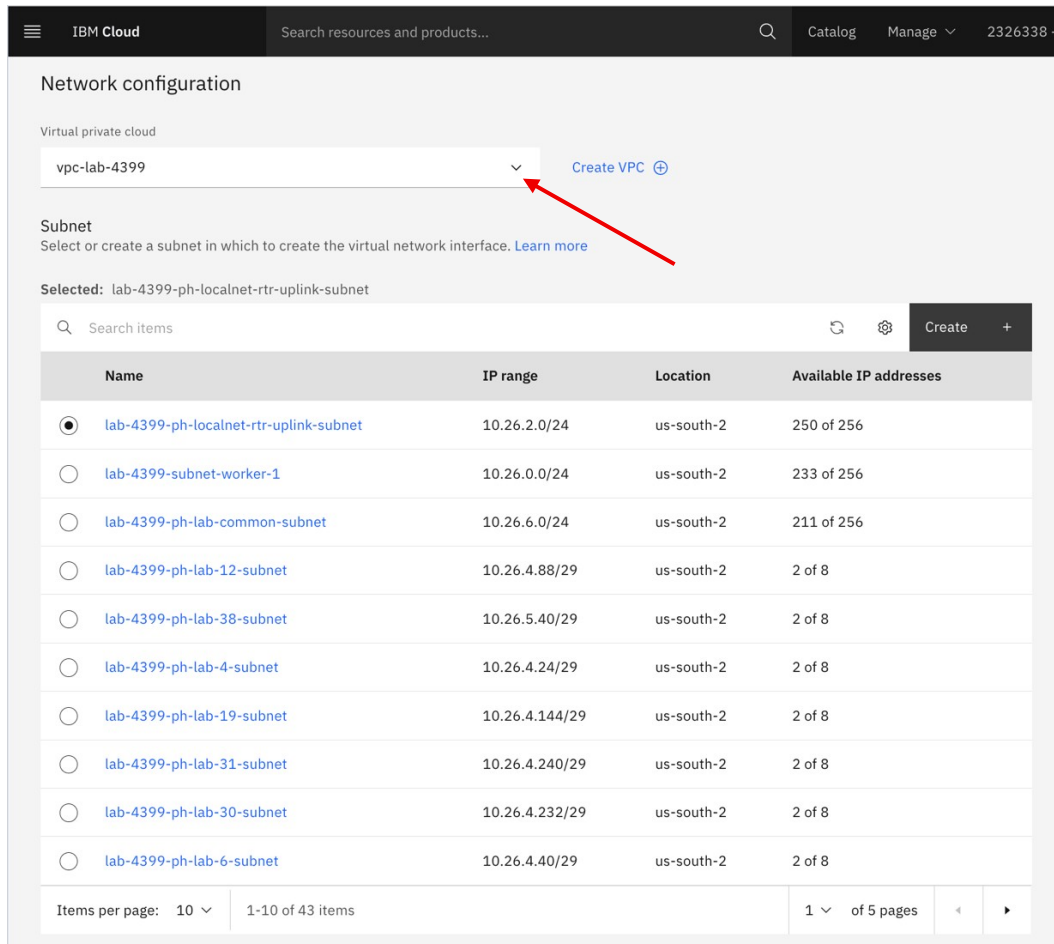
Resource group ⓘ
 [Create resource group](#) +

[View all resource groups](#) ⓘ

Tags (optional) ⓘ

Access management tags (optional) ⓘ

b. Scroll down to **Network configuration**. Confirm the VPC is **vpc-lab-4399**. Then, choose the subnet that corresponds with **YOUR lab number**. (e.g. Lab 1 would select lab-4399-ph-lab-1-subnet)



Network configuration

Virtual private cloud

vpc-lab-4399 ▼ [Create VPC](#)

Subnet

Select or create a subnet in which to create the virtual network interface. [Learn more](#)

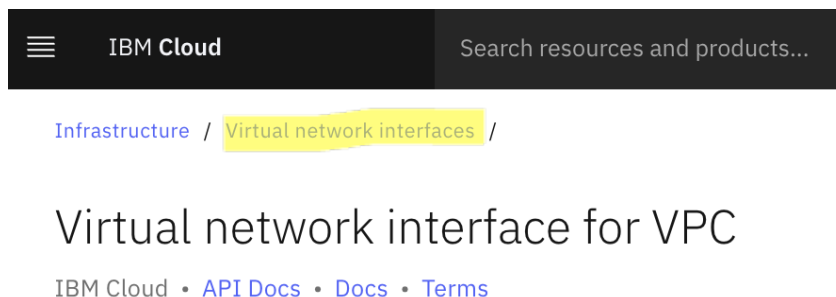
Selected: lab-4399-ph-localnet-rtr-uplink-subnet

Search items [Create](#)

Name	IP range	Location	Available IP addresses
☒ lab-4399-ph-localnet-rtr-uplink-subnet	10.26.2.0/24	us-south-2	250 of 256
☐ lab-4399-subnet-worker-1	10.26.0.0/24	us-south-2	233 of 256
☐ lab-4399-ph-lab-common-subnet	10.26.6.0/24	us-south-2	211 of 256
☐ lab-4399-ph-lab-12-subnet	10.26.4.88/29	us-south-2	2 of 8
☐ lab-4399-ph-lab-38-subnet	10.26.5.40/29	us-south-2	2 of 8
☐ lab-4399-ph-lab-4-subnet	10.26.4.24/29	us-south-2	2 of 8
☐ lab-4399-ph-lab-19-subnet	10.26.4.144/29	us-south-2	2 of 8
☐ lab-4399-ph-lab-31-subnet	10.26.4.240/29	us-south-2	2 of 8
☐ lab-4399-ph-lab-30-subnet	10.26.4.232/29	us-south-2	2 of 8
☐ lab-4399-ph-lab-6-subnet	10.26.4.40/29	us-south-2	2 of 8

Items per page: 10 ▼ 1-10 of 43 items 1 ▼ of 5 pages

- b. Leave all remaining fields as their defaults. **DO NOT CLICK CREATE.** Navigate back to the **Virtual network interfaces** page by clicking the tab at the top of the page.



IBM Cloud Search resources and products...

Infrastructure / Virtual network interfaces /

Virtual network interface for VPC

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1. Investigate the VNI

- a. Find the VNI that corresponds with **YOUR lab number** in the list. Click on it.

Hint: It should be in the format of lab-4399-ph-lab-<your lab number>-vm-1

Virtual network interfaces for VPC

Improve the capability and flexibility of network interfaces in your VPC with enhanced VPC services integration, high-availability, multiple IP addresses, and transferable interfaces. [Learn more](#)

Region: Dallas (us-south) Search items

Name

Status

Resource group

Virtual private cloud

Subnet

Primary IP

Floating IPs

Target resource

lab-4399-ph-layer2-routed-1-uplink-vni

Stable

techexchange-2026-lab-4399

vpc-lab-4399

lab-4399-ph-localnet-rtr-uplink-subnet

10.26.2.4

0

—

:

lab-4399-ph-lab-9-vm-1

Stable

techexchange-2026-lab-4399

vpc-lab-4399

lab-4399-ph-lab-9-subnet

10.26.4.68

0

—

:

lab-4399-ph-lab-common-lab-3

Stable

techexchange-2026-lab-4399

vpc-lab-4399

lab-4399-ph-lab-common-subnet

10.26.6.43

0

—

:

lab-4399-ph-lab-23-vm-1

Stable

techexchange-2026-lab-4399

vpc-lab-4399

lab-4399-ph-lab-23-subnet

10.26.4.180

0

—

:

lab-4399-ph-lab-common-lab-6

Stable

techexchange-2026-lab-4399

vpc-lab-4399

lab-4399-ph-lab-common-subnet

10.26.6.42

0

—

:

lab-4399-ph-lab-common-lab-19

Stable

techexchange-2026-lab-4399

vpc-lab-4399

lab-4399-ph-lab-common-subnet

10.26.6.41

0

—

:

lab-4399-ph-lab-common-lab-34

Stable

techexchange-2026-lab-4399

vpc-lab-4399

lab-4399-ph-lab-common-subnet

10.26.6.40

0

—

:

lab-4399-ph-lab-19-vm-1

Stable

techexchange-2026-lab-4399

vpc-lab-4399

lab-4399-ph-lab-19-subnet

10.26.4.148

0

—

:

lab-4399-ph-lab-36-vm-1

Stable

techexchange-2026-lab-4399

vpc-lab-4399

lab-4399-ph-lab-36-subnet

10.26.5.28

0

—

:

lab-4399-ph-lab-28-vm-1

Stable

techexchange-2026-lab-4399

vpc-lab-4399

lab-4399-ph-lab-28-subnet

10.26.4.220

0

—

:

Items per page: 10 1-10 of 90 items

1 of 9 pages

- b. Attach the VNI to the Worker Node
 - a. Click the **Hamburger menu**.
 - b. Navigate to **Infrastructure** → **OpenShift virtualization** → **virtualization-4399**.
 - c. In the left navigation click **VNI attachments**.
 - d. Scroll until you find an attached VNI with **your lab number**.
 - e. Open it and navigate around.

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Search resources and products...

Q Catalog Manage

2326338 - ITZ-VMWARE

?

Infrastructure / OpenShift Virtualization clusters /

virtualization-4399 Normal [Add tags](#)

Help [OpenShift web console](#) [Actions](#)

Overview

Worker nodes

Worker pools

Networking

VNI attachments

Ingress

Virtual network interface attachments

Subnet: Filter...

Worker node: Filter...

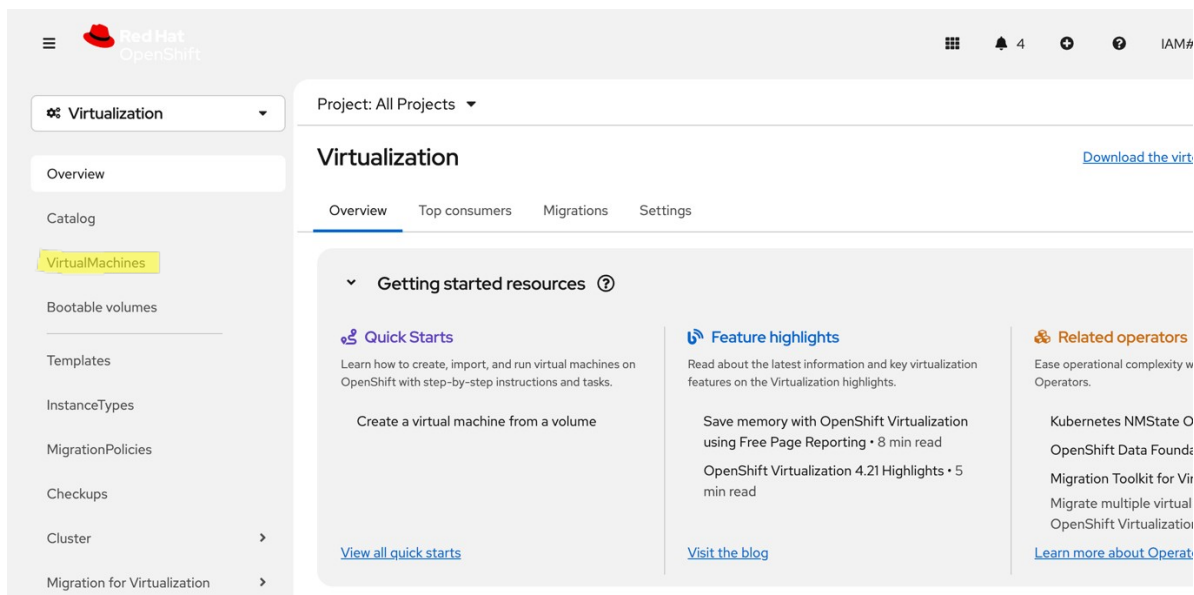
Q Search

Attach VNIs +

☐	VNI name	Worker node	Can float	Subnet	VLAN ID	Primary IP
☐	lab-4399-ph-lab-1-vm-1	0000041f	Yes	lab-4399-ph-lab-1-subnet	400	10.26.4.4
☐	lab-4399-ph-lab-10-vm-1	00000141	Yes	lab-4399-ph-lab-10-subnet	409	10.26.4.76
☐	lab-4399-ph-lab-11-vm-1	0000034d	Yes	lab-4399-ph-lab-11-subnet	410	10.26.4.84
☐	lab-4399-ph-lab-12-vm-1	0000041f	Yes	lab-4399-ph-lab-12-subnet	411	10.26.4.92
☐	lab-4399-ph-lab-13-vm-1	00000141	Yes	lab-4399-ph-lab-13-subnet	412	10.26.4.100
☐	lab-4399-ph-lab-14-vm-1	00000141	Yes	lab-4399-ph-lab-14-subnet	413	10.26.4.108
☐	lab-4399-ph-lab-15-vm-1	00000141	Yes	lab-4399-ph-lab-15-subnet	414	10.26.4.116
☐	lab-4399-ph-lab-16-vm-1	0000041f	Yes	lab-4399-ph-lab-16-subnet	415	10.26.4.124
☐	lab-4399-ph-lab-17-vm-1	0000041f	Yes	lab-4399-ph-lab-17-subnet	416	10.26.4.132
☐	lab-4399-ph-lab-18-vm-1	00000141	Yes	lab-4399-ph-lab-18-subnet	417	10.26.4.140
☐	lab-4399-ph-lab-19-vm-1	0000041f	Yes	lab-4399-ph-lab-19-subnet	418	10.26.4.148
☐	lab-4399-ph-lab-2-vm-1	0000024f	Yes	lab-4399-ph-lab-2-subnet	401	10.26.4.12
☐	lab-4399-ph-lab-20-vm-1	0000034d	Yes	lab-4399-ph-lab-20-subnet	419	10.26.4.156

2. Navigate to VirtualMachines.

- Switch to the **OpenShift web console** tab. (If needed, navigate back to step 0 to see how to do so).
- In the left navigation menu, click the dropdown, **Virtualization** → **VirtualMachines**. Leave the Project selector set to **All Projects**.



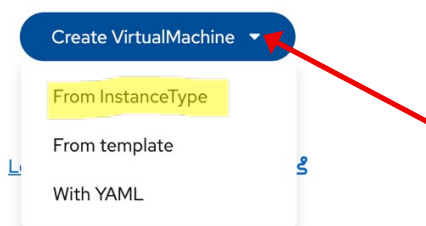
3. Open the VM Creation Wizard

- Click **Create VirtualMachines** → **From InstanceType**.



No VirtualMachines found

Click **Create VirtualMachine** to create your first VirtualMachine or view the [catalog](#) tab to create a VirtualMachine from the available options



4. Select the Boot Volume

- Under the **InstanceTypes** tab, select **centos-stream9** as the OS.

Project: default ▾

Create new VirtualMachine

Select an option to create a VirtualMachine from.

InstanceTypes Template catalog

1 Select volume to boot from ⓘ

Add volume

Volumes project

PR All projects ▾

Filter ▾

Search by name...

1 - 8 of 9 ▾

< >

Show all

★ ↑	Volume name ↓	Architecture ↓...	Namespace ↓	Operating system ↓	Storage class ↓	Size ↓	Description ↓
★	centos-stream9	-	openshift-virtualization-os-images	CentOS Stream 9	ocs-storagecluster-rbdplugin-snapclass	30 GiB	-
★	win2k22	-	openshift-virtualization-os-images	Microsoft Windows Server 2022 (virtio)	ocs-storagecluster-ceph-rbd-virtualization	60 GiB	-
★	fedora	-	openshift-virtualization-os-images	Fedora (amd64)	ocs-storagecluster-rbdplugin-snapclass	30 GiB	-
★	rhel10	-	openshift-virtualization-os-images	Red Hat Enterprise Linux 10 (amd64)	ocs-storagecluster-rbdplugin-snapclass	30 GiB	-
★	win2k25	-	openshift-virtualization-os-images	Microsoft Windows Server 2025 (virtio)	ocs-storagecluster-ceph-rbd-virtualization	64 GiB	-

5. Select the InstanceType

- Choose the **General Purpose** card (U series), then from its dropdown select **small: 1 CPUs, 2 GiB Memory**.

★ win11

images

(amd64)

snapclass

openshift-virtualization-os-images

Microsoft Windows 11 (virtio)

ocs-storagecluster-ceph-rbd-virtualization

64 GiB

-

2 Select InstanceType

Red Hat provided User provided

Network
N series ▾

Overcommitted
O series ▾

Memory Intensive
M series ▾

Realtime
RT series ▾

Dedicated vCPU
D series ▾

Compute Exclusive
CX series ▾

U series ▾

nano: 1 CPUs, 512 MiB Memory

micro: 1 CPUs, 1 GiB Memory

small: 1 CPUs, 2 GiB Memory ✓

medium: 1 CPUs, 4 GiB Memory

2xmedium: 2 CPUs, 4 GiB Memory

large: 2 CPUs, 8 GiB Memory

xlarge: 4 CPUs, 16 GiB Memory

2xlarge: 8 CPUs, 32 GiB Memory

4xlarge: 16 CPUs, 64 GiB Memory

8xlarge: 32 CPUs, 128 GiB Memory

6. Name the VM

- In the **VirtualMachine details** section, set the **Name** to **<initials>-localnet-vm** (e.g. **js-localnet-vm**).

DO NOT CLICK Create VirtualMachine. Click **Customize VirtualMachine** – the default wizard puts the VM on the pod network with no fixed MAC. You need to set the Localnet network and MAC address before creation.

3 VirtualMachine details

Name *

js-layer2-vm

Operating system

CentOS Stream 9

InstanceType

u1.small

CPU | Memory

1 CPU | 2 GiB Memory

Project

lab-4399-ph-lab-1

Disk size

- 30 + GiB

Storage class

ocs-storagecluster-ceph-rbd-virtu...

Public SSH key

Not configured

Dynamic SSH key injection ?

Start this VirtualMachine after creation

Create VirtualMachine

Customize VirtualMachine

Cancel

View YAML & CLI

7. Open Configuration

- In the Customize screen, click the **Configuration** tab, then navigate to the **Network** tab on the left. Click the kebab menu (:) on the **default** interface row and choose **Edit**.

YAML Configuration

Details

Storage

Network

Scheduling

SSH

Initial run

Network interfaces

Add network interface

Filter

Name

Search by name...

Name	Model	Network	State	Type	MAC addr...
default	-	Pod networking		Masquerade	-

Set link down

Edit

Delete

Start this VirtualMachine after creation

Create VirtualMachine

Cancel

8. Set Network and MAC Address

a. In the **Edit network interface** dialog:

- **Network:** Choose the one that matches **YOUR VM** (e.g. **lab-4399-ph-lab-1**)
- Expand **Advanced Settings** → set **MAC address** to the MAC Address you found in your VNI.

Edit network interface ✕

Name *

default

Model

virtio

Network * ⓘ

lab-4399-ph-lab-1/lab-4399-ph-lab-1

▼ Advanced Settings

MAC address

Link state

Up

Save Cancel

Click **Save**.

9. Set the Login Password

- In the left-hand list, click **Initial run**, then click **Edit** on the Cloud-init section.

YAML Configuration

Q

Details

Storage

Network

Scheduling

SSH

Initial run

Metadata

Initial run

Cloud-init [Edit](#)

You can use cloud-init to initialize the operating system with a specific configuration when the VirtualMachine is started. The cloud-init service is enabled by default in Fedora and RHEL images. [Learn more](#)

User	Password	Network data
centos	*****	Default

Sysprep [Windows only](#) [Edit](#)

Sysprep is an automation tool for Windows that automates Windows setup, and custom software provisioning. An answer file is an XML-based file that contains setting definitions and values to use during Windows Setup [Learn more](#)

Not available

- Set the password to **ChangeMe123!** and click **Apply**.

Cloud-init ✕

You can use cloud-init to initialize the operating system with a specific configuration when the VirtualMachine is started. The cloud-init service is enabled by default in Fedora and RHEL images. [Learn more](#)

Configure via: ☒ Form view ☐ Script

User *

centos

Password

ChangeMe123!

Password for this username - [generate password](#)

☐ Add network data

check this option to add network data section to the cloud-init script.

[Apply](#) [Cancel](#)

10. Create the VM

- Scroll down and click the blue **Create VirtualMachine** button at the bottom of the screen.

☒ Start this VirtualMachine after creation



11. Watch the VM Boot

- Return to the VirtualMachine list. Watch the Status column progress:

Provisioning → Starting → Running

VM js-layer2-vm Running

[Overview](#) [Metrics](#) [YAML](#) [Configuration](#) [Events](#) [Console](#) [Snapshots](#) [Diagnostics](#)

[Details](#)

Name	js-layer2-vm amd64	VNC console Open web console  
Status	 Running	
Created	1 minute ago (1 minute ago)	
Operating system	CentOS Stream 9	
CPU Memory	1 CPU 2 GiB Memory	
Time zone	EDT	
InstanceType	CR u1.small	
Preference	CR centos.stream9	
Hostname	js-layer2-vm	

12. Inspect the VM Overview

- a. Scroll down and look at the **Network** tile – it shows the static IP assigned to this VM.

Network (!)	
Name	IP address
default	172.17.2.13
Internal FQDN	js-layer2-vm.headless.l...

- b. Click the **Network** tile header to jump to **Configuration → Network**. Point out:
- Network attachment: **Pod Networking**.
 - MAC address matches the VNI reservation – this is what pins the IP permanently.

Details

Storage

Network

Scheduling

SSH

Initial run

Metadata

Network interfaces

Add network interface

Filter

Name

Search by name...

Name	Model	Network	State	Type	MAC ad...
default	virtio	Pod networking		Masquerade	02:a1:c3:6e:18:89

13. Open the In-Browser Console

- a. Click the **Console** tab. Log in as **Centos** with the password as **ChangeMe123!**

IBM TechXchange

*Hint: Use the **Copy** and **Paste to Console** buttons to paste logon credentials.*

VM js-layer2-vm Running

Overview Metrics YAML Configuration Events **Console** Snapshots Diagnostics

Guest login credentials ⓘ User name Password ⓘ

Paste to console en-us VNC console Send key Disconnect

```
CentOS Stream 9
Kernel 5.14.0-749.el9.x86_64 on an x86_64
Activate the web console with: systemctl enable --now cockpit.socket
js-layer2-vm login: _
```

14. Show the Static IP Inside the VM

- In the VM console, run:

```
ip addr show
```

```
CentOS Stream 9
Kernel 5.14.0-749.el9.x86_64 on an x86_64

Activate the web console with: systemctl enable --now cockpit.socket

js-layer2-vm login: centos
Password:
[centos@js-layer2-vm ~]$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 8900 qdisc fq_codel state UP group default qlen 1000
    link/ether 02:a1:c3:6e:18:89 brd ff:ff:ff:ff:ff:ff
    altname enp1s0
    inet 10.0.2.2/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 86312281sec preferred_lft 86312281sec
    inet6 fe80::a1:c3ff:fe6e:1889/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[centos@js-layer2-vm ~]$ s
```

b. Then run:

```
curl http://lab.techzone.ibm.local/
```

Congratulations, you have completed part 1 of this lab!

Migration Toolkit for Virtualization – Migrate a VMware VM to ROVS

This end-to-end lab takes you from a clean ROVS cluster to a successfully migrated VMware VM – all in one walkthrough. It ties together the MTV operator (pre-installed on ROVS), VPC networking, and ODF storage into a

IBM TechXchange

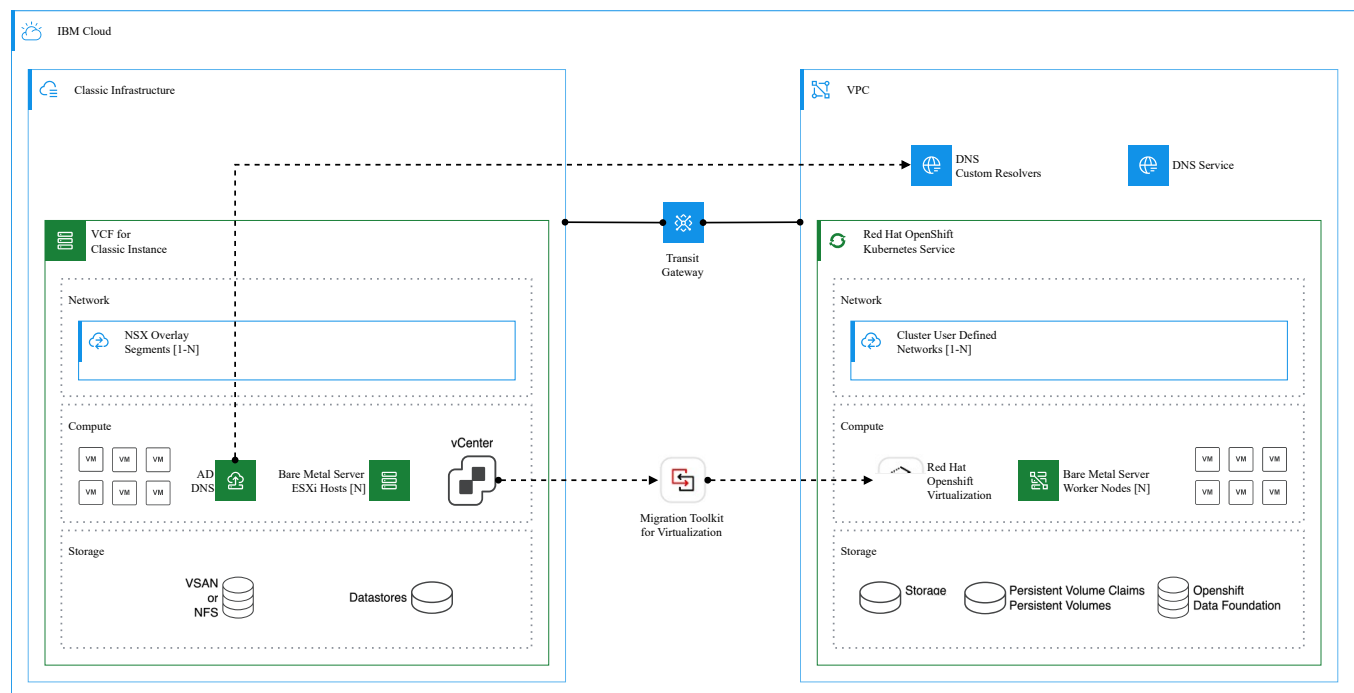
complete migration scenario. MTV migrates VMs to Red Hat OpenShift Virtualization running on Red Hat OpenShift.

What MTV Does:

1. **Reads** the VM's disk, CPU, memory, and network configuration from vCenter.
2. **Converts** the VM disk from VMDK format to a KubeVirt-compatible image using virt-v2v.
3. **Create** a VirtualMachine CR in OpenShift with mapped resource.
4. **Starts** the VM and (optionally) powers off the source.

Cold Migration – The source virtual servers are stopped while the data is copied. The default migration.

Warm Migration – Most of the data during the precopy stage is copied. Then, the virtual servers are stopped, and the remaining data is copied during the cutover stage.

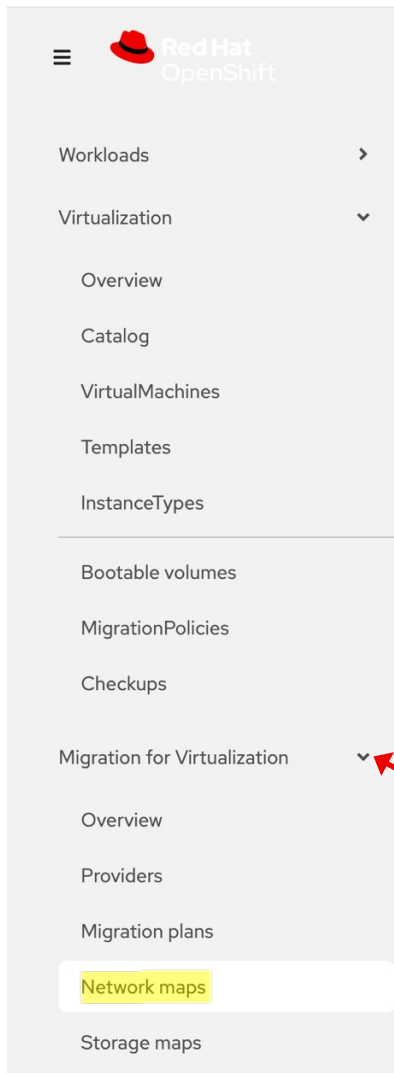


Cold Migration

1. Create a Network map

- a. Open the **OpenShift web console**. In the left navigation menu, click **Migration for Virtualization** → **Network maps**.

IBM TechXchange



- b. Click **Create network map** → **Create with form**.

Network maps [?]

[Tips and tricks](#)[Create network map](#) ▼

Status ▼

Name ▼

🔍 Filter by name



🔧 Manage columns

Name ↑

Status ↓

Source provider ↓

Target provider ↓

Owner ↓



No network maps found in project **lab-4399-ph-lab-1**

Migration network maps are used to map networks between source and target providers.

[Create network map](#) ▼[Create with form](#)[Create with YAML](#)

c. Fill out the form:

- **Network map name:** <your lab number>-networkmap (e.g. lab1-networkmap)
- **Project:** Select YOUR project
- **Source provider:** ocp-gym-vc
- **Target provider:** host
- **Source network:** 6a7c55c9e78638eb8066f39-segment
- **Target network:** lab-4399-ph-lab-<your lab number>/lab-4399-ph-lab-<your lab number>

Create network map

Network map name *

lab1-networkmap-js

Project * ⓘ

lab-4399-ph-lab-1

Source provider *

ocp-gym-vc

Target provider *

host

Source network *

6a7c556c9e78638eb8066f39-se... ▼

Target network *

lab-4399-ph-lab-1/lab-4399-ph... ▼

+ [Add mapping](#)

Create

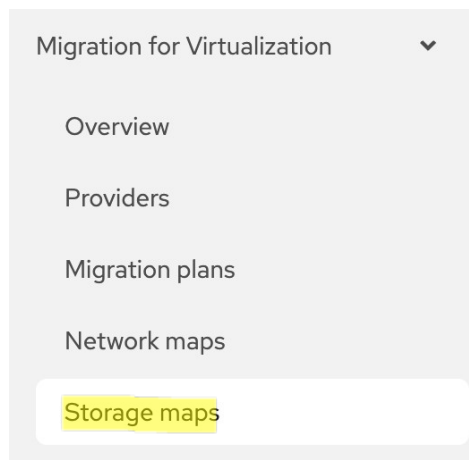
Cancel

d. Click **Create**.

2. Create a Storage map

IBM TechXchange

- a. In the left navigation menu, click **Migration for Virtualization** → **Storage maps**.



- b. Click **Create storage map** → **Create with form**.

Storage maps ⓘ

[Tips and tricks](#)

Create storage map ▾

Status ▾

Name ▾

🔍 Filter by name



🔧 Manage columns

Name ↑

Status ↓

Source provider ↓

Target provider ↓

Owner ↓



No storage maps found in project lab-4399-ph-lab-2.

Migration storage maps are used to map storages between source and target providers.

Create storage map ▾

Create with form

Create with YAML

- c. Fill out the form:

- **Storage map name:** <your lab number>-storagemap (e.g. lab1-storagemap)

IBM TechXchange

- **Project:** Select YOUR project
- **Source provider:** ocp-gym-vc
- **Target provider:** host
- **Source network:** 6a7c55c9e78638eb8066f39-storage
- **Target network:** ocs-storagecluster-ceph-rpd

Create storage map

Storage map name *

lab1-storagemap

Project * ⓘ

lab-4399-ph-lab-1 ▼

Source provider *

ocp-gym-vc ▼

Target provider *

host ▼

Source storage *

6a7c556c9e78638eb8066f3... ▼

Target storage *

ocs-storagecluster-ceph-rb... ▼



+ [Add mapping](#)

Create

Cancel

d. Click **Create**.

3. Access the MTV UI

- a. In the left navigation menu, click **Migration for Virtualization** → **Migration plans**.

VirtualMachines

Bootable volumes

Templates

InstanceTypes

MigrationPolicies

Checkups

Cluster

Migration for Virtualization

Storage migrations

Overview

Providers

Migration plans

Network maps

Storage maps

Project: All Projects

Migration plans

Filter by name

Show archived

Manage columns

Name

Project

Source prov...

Target proj...

Vir...

Mi...

Mi...

Mi...

No plans found

Migration plans are used to document the moving of virtualization workloads from source providers to target providers.

Create plan

b. Click **Create Plan**.

4. Create a Migration plan

In the General tab, filling in the following:

- Plan name:** cold-migration-lab<your lab number> (e.g. cold-migration-lab1)
- Plan project:** Select YOUR project number (e.g. lab-4399-ph-lab-1)

IBM TechXchange

- c. **Source provider:** vcenter-prod
- d. **Target provider:** host
- e. **Target namespace:** vm-lab<your lab number> (e.g. vm-js)

Create migration plan

[Tips and tricks](#)

1 Basic setup ▾

General

Virtual machines

Network map

Storage map

2 Additional setup ▶

3 Review and create

General

Plan information

Name your plan and choose the project you would like it to be created in.

Plan name *

Plan project * ⓘ

Description

Source and target providers

Select the provider you would like to migrate your virtual machines from (source provider) and the provider you want to migrate your virtual machines to (target provider).

Source provider *

Target provider *

Back

Next

Cancel

Click **Next**.

3. Select your project's VM

1. Expand the **lab-4399** dropdown.
2. Scroll to find **YOUR lab number** → click the **check box** next to your lab number.

Create migration plan

[Tips and tricks](#)

1 Basic setup

General

Virtual machines

Network map

Storage map

Migration type

2 Additional setup

3 Review and create

Virtual machines

Select the virtual machines you want to migrate. To help find the virtual machines you're looking for, try using the filters.

1 selected

VM name

Filter by VM name

Manage columns

All

Selected

1 - 10 of 43

<<

<

1

>

>>

Name	C...	Ins...	Host
> ☐ 6a7c556c9e78638eb8066f39			2 VMs
> ☒ lab-4399			40 VMs
> ☒ lab-1	1	Not inspected	ocp-gym011.techzone.ibm.local
> ☐ lab-10	1	Not inspected	ocp-gym028.techzone.ibm.local
> ☐ lab-11	1	Not inspected	ocp-gym029.techzone.ibm.local

Back

Next

Cancel

3. Click **Next**.

4. Network Map

- Confirm "Use an existing network map" is selected.
- From the dropdown, select your networkmap from the dropdown.

Create migration plan

1 Basic setup ▾

General

Virtual machines

Network map

Storage map

Migration type

2 Additional setup >

Network map

Select an existing network map or use a new network map.

☒ Use an existing network map

Existing network map options are limited to those without an owner reference. Upon creation of this plan, a new network map will be created with this plan as its owner.

Network map *

Select network map ⓘ ▾

ⓘ Network map is required.

☐ Use new network map

Use the suggested network mapping and add mappings to it, or create a brand new one as needed. A new map, with this plan as its owner, will be automatically created based on your selected mappings.

c. Click **Next**.

5. Storage Map

- Confirm “Use an existing storage map” is selected.
- From the dropdown, select your storage map

Create migration plan

1 Basic setup ▾

General

Virtual machines

Network map

Storage map

Migration type

2 Additional setup >

Storage map

Select an existing storage map or use a new storage map.

☒ Use an existing storage map

Existing storage map options are limited to those without an owner reference. Upon creation of this plan, a new storage map will be created with this plan as its owner.

Storage map *

Select storage map ⓘ ▾

ⓘ Storage map is required.

☐ Use new storage map

Use the suggested storage mapping and add mappings to it, or create a brand new one as needed. A new map, with this plan as its owner, will be automatically created based on your selected mappings.

Click **Next**.

6. Select Migration Type

- Select **Cold migration**.
- Click **Next**.

Create migration plan

1 Basic setup ▾

General

Virtual machines

Network map

Storage map

Migration type

Migration type

☒ Cold migration

A cold migration moves a shut-down VM between hosts.

☐ Warm migration ?

A warm migration moves an active VM with minimal downtime (not zero-downtime).

7. Create Plan

- Leave all other settings how they are.
- Click **Skip to Review**.
- Click **Create plan**.

Back

Create plan

8. Start the Migration

- Back on **Migration plans** → click the **3 dots (:)** on your migration plan.
- Click **Start**.



9. Monitor the migration progress.

- a. Click on the migration plan. Wait for the status to reach **Completed**.

Warm Migration

1. Access the MTV UI

- a. Open the **OpenShift web console**.
- b. In the left navigation menu, click the dropdown and navigate to **Migration for Virtualization** → **Migration plans**.
 - i. You will see your previous cold migration in the list.
- c. Click **Create plan**.

Migration plans ⓘ

[Tips and tricks](#)[Create plan](#)

Name ▾

🔍 Filter by name

➔

☐ Show archived

 Manage columns

Name ↑	Source pro... ↓	Target project ↓	Vi... ↓	Migrati... ↓	M... ↓	Migration started ↓	
PL test	vm ocp-gym-vc	PR lab-4399-ph-lab-1	 1VM	Complete  1	Cold	 Sep 29, 2026, 8:44 AM	

2. On the General settings page, fill in the following fields:

- a. **Plan name:** warm-migration-<initials> (e.g. warm-migration-js)
- b. **Plan project:**
- c. **Source provider:** vcenter-prod
- d. **Target provider:** host
- e. **Target namespace:** vm-<initials> (e.g. vm-js)

Create migration plan

[Tips and tricks](#)

1 Basic setup ▾

General

Virtual machines

Network map

Storage map

2 Additional setup ►

3 Review and create

General

Plan information

Name your plan and choose the project you would like it to be created in.

Plan name *

Plan project * ⓘ

Description

Source and target providers

Select the provider you would like to migrate your virtual machines from (source provider) and the provider you want to migrate your virtual machines to (target provider).

Source provider *

Target provider *

Back

Next

Cancel

Click **Next**.

3. Select the **Source VM**.

Create migration plan

Tips and tricks

1 Basic setup ▾

General

Virtual machines

Network map

Storage map

Migration type

2 Additional setup ▶

3 Review and create

Virtual machines

Select the virtual machines you want to migrate. To help find the virtual machines you're looking for, try using the filters.

☐ 1 selected ▾

▼ VM name ▾

🔍 Filter by VM name

🗲 Manage columns

All Selected

1 - 10 of 43 ▾ << < 1 of 5 > >>

Name ↑	Conce... ↑	Ins... ↑	🔍	Host ↑
▼ 📁 📁 vm				

Back

Next

Cancel

Click **Next**.

5. Select your **Network** map.

Create migration plan

[Tips and tricks](#)

1 Basic setup ▾

General

Virtual machines

Network map

Storage map

Migration type

2 Additional setup >

3 Review and create

Network map

Select an existing network map or use a new network map.

☒ Use an existing network map

Existing network map options are limited to those without an owner reference. Upon creation of this plan, a new network map will be created with this plan as its owner.

Network map *

Select network map ▾

☐ Use new network map

Use the suggested network mapping and add mappings to it, or create a brand new one as needed. A new map, with this plan as its owner, will be automatically created based on your selected mappings.

Back

Next

Cancel

Click **Next**.

6. Select your **Storage map**.

Create migration plan

[Tips and tricks](#)

1 Basic setup ▾

General

Virtual machines

Network map

Storage map

Migration type

2 Additional setup ►

3 Review and create

Storage map

Select an existing storage map or use a new storage map.

☒ Use an existing storage map

Existing storage map options are limited to those without an owner reference. Upon creation of this plan, a new storage map will be created with this plan as its owner.

Storage map *

Select storage map ▾

☐ Use new storage map

Use the suggested storage mapping and add mappings to it, or create a brand new one as needed. A new map, with this plan as its owner, will be automatically created based on your selected mappings.

Back

Next

Cancel

Click **Next**.

7. Select **Warm migration**.

Create migration plan

Tips and tricks

1 Basic setup ▾

General

Virtual machines

Network map

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Migration type

2 Additional setup ▶

3 Review and create

Migration type

☐ Cold migration

A cold migration moves a shut-down VM between hosts.

☒ Warm migration ⓘ

A warm migration moves an active VM with minimal downtime (not zero-downtime).

Back

Next

Skip to review

Cancel

Click **Skip to Review**.

8. Click **Create plan**.

Create migration plan

[Tips and tricks](#)

- 1 Basic setup >
- 2 Additional setup >
- 3 Review and create

Review and create

Make sure your migration plan looks correct. To make any changes, click directly on the step you want to go to or press the back button.

General

[Edit step](#)

Plan name warm-migration

Plan project

Description

Source provider source-vsphere

Target provider host

Target project default

Virtual machines

[Edit step](#)

Selected VMs [1 virtual machine selected](#)

Network map

[Edit step](#)

Network map rovs-localnet-map

[Back](#)[Create plan](#)[Cancel](#)

9. Start the Migration.

Back on **Migration plans** → click the **3 dots (:)** on your migration plan. Click **Start**.



10. Monitor the migration progress.

Click on the migration plan. Wait for the status to reach **Completed**.

